

Sprint 1 Feature Documentation

Sprint: Sprint 1

Product: News 4 U

Team: *Agile Alligators*

Sprint Goal: Deliver a stable, end-to-end foundation that supports content aggregation, user management, and future AI personalization.

1. Summary of Sprint Outcomes

During this sprint, the team delivered a **fully functional baseline platform** that establishes the core system architecture and user experience. The focus was on enabling reliable user access, ingesting real article data, and building a scalable foundation that supports future AI-driven capabilities.

The team delivered authenticated access, a reliable content ingestion pipeline, and a functional user interface that enables real-world usability testing. All completed work aligns directly with predefined product backlog items.

2. Completed Features & User Story Mapping

User Management & Authentication

Description: Implemented secure user onboarding and access control to enable personalized experiences and protect system resources.

Completed PBIs:

- **Register Account** — *As a new user, I want to register an account with my email and password, so that I can access features.*
- **Login** — *As a returning user, I want to log in with my email and password, so that I can access my account.*

Users can authenticate securely and access logged-in functionality.

Content Aggregation Pipeline

Description: Built the backend pipeline to fetch, store, and process real news articles from external sources.

Completed PBIs:

- **Fetch News Sources** — *As a user, I want articles fetched from multiple news sources so that I can access diverse perspectives.*
- **Categorize Content** — *As a user, I want articles tagged by topic so I can browse relevant news.*
- **Timestamp Articles** — *As a user, I want to see publication timestamps so I know how current an article is.*

The backend currently ingests article metadata from news APIs (e.g., title, source, timestamps), which is processed and rendered in the frontend. Persistent storage of full article content via scraping will be implemented in a future sprint.

Frontend User Interface

Description: Translated wireframes into working UI components covering the full end-to-end user experience, including authentication, content discovery, article reading, and interaction flows. The frontend is implemented for all major user-facing features and enables early usability validation of complex and high-risk interactions prior to full backend completion.

Completed / Partially Implemented PBIs (UI-level):

- **Register Account** — user onboarding and registration screens
- **Login** — authentication flow and error states
- **Browse Articles / Trending Topics** — headline lists and topic-based exploration
- **Highlight Technical Terms (UI-level)** — interactive highlights within articles
- **Provide Explanations (UI-level)** — pop-up simplification or definition overlays
- **Feedback on Tailored Articles (UI-level)** — feedback controls at the end of articles

The UI supports usability testing of high-risk flows, including login and signup error recovery, navigation transitions, article reading interactions, AI-triggered pop-ups, and feedback submission, even where underlying intelligence or data ingestion is still being finalized.

Database & Infrastructure Setup

Description: Set up core infrastructure to support persistence, scalability, and clean integration.

Completed PBIs:

- *Infrastructure enabler stories* supporting:
 - User profiles
 - Article storage
 - Topic tagging

Backend and frontend communicate through authenticated APIs.

3. Architecture & Design Alignment

The implemented system follows the original architectural design with clear separation of concerns:

- **Frontend:** UI and user interaction logic
- **Backend services:** Authentication, content ingestion, API orchestration
- **Database:** Yugabyte for structured and scalable data storage
- **External services:** News APIs supplying article content

Key Design Decisions

- Prioritized end-to-end data flow before advanced AI features.
- Kept architecture modular to reduce technical risk in later sprints.
- Ensured UI flows were in place to support early usability testing and validation.

These decisions reduce rework and make future AI integration more predictable.

4. Known Limitations

The following capabilities are intentionally deferred:

- No LLM-based article simplification yet
- Recommendation logic is rule-based or placeholder
- Feedback loops do not yet influence system behavior

These were scoped out of this sprint to focus on building a dependable platform foundation.

5. Sprint Conclusion

Sprint 1 successfully delivered all **core functional PBIs** required to support future AI-driven value. The system is stable, usable, and architecturally aligned with the original design, enabling Sprint 2 to focus on intelligence, personalization, and quality improvements.